EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Publishing every lesson guarantees each student is ready for assessment.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

DO NOW — RECALL

Final mixed-paper readiness

Name: _____ Class: _____ Date: _____

1. Difference between MAR and MDR?

2. How to encode a negative integer in two's complement?

3. What does $O(n^2)$ describe?

4. Full adder A=1 B=1 carry-in=1. State sum and carry.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

MAR holds address; MDR transports data/instruction contents.

2. Model answer

Invert fixed-width positive representation then add one.

3. Model answer

Quadratic asymptotic growth.

4. Model answer

$1+1+1=3=11_2$; sum1 carry1.

By the end, you can…

- Apply Paper 1 and Paper 2 knowledge together
- Demonstrate precise independent working
- Identify the strongest remaining revision priority

Vocabulary — know these terms

readiness — Ability to apply without prompts.

synoptic — Knowledge across multiple topics.

method evidence — Reasoning supporting an answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Q3 [9]: Evaluate cloud storage versus USB-only storage for an inclusive school revision service; weigh access, reliability, security. [9]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Q1 [8]: In a simplified sequential fetch, PC=12 and memory address 12 stores instruction ADD 7. State MAR, MDR, CIR and next PC (assume no branch or interrupt). [4] Convert −5 to 8-bit two's complement. [4]

Question 2

Q2 [8]: Trace binary search for 11 in [1,3,5,7,9,11,13]. [4] Explain base case and recursion progress. [4]

NEW LEARNING

Read, annotate and explain

Identify

Retrieve knowledge across both components

Model

Show working in numeric and algorithm tasks

Apply

Check terminology and boundary cases

Evaluate

Prioritise a final independent improvement target

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Normalise positive $0.0101_2 \times 2^3$.

Explain the method

Shift mantissa one position left and reduce exponent by one.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve